Employee Portal – IT Inventory Management & Ticketing System

Entity Relationship Diagram (ERD) · Fauji Foods Limited / Fauji Fresh n Freeze · Crow's Foot Notation

A · ORGANIZATION & ACCESS MANAGEMENT

roles

PK	role_id	INT
	role_name	VARCHAR(50) UQ
	description	VARCHAR(255)

users

PK	user_id	INT
	username	VARCHAR(50) UQ
	email	VARCHAR(100) UQ
	password_hash	VARCHAR(255)
FK	role_id	INT
	is_active	BOOLEAN
	created_at	DATETIME
	last_login	DATETIME NULL

employees

PK	employee_id	INT
FK	user_id	INT UQ
	employee_code	VARCHAR(20) UQ
	full_name	VARCHAR(100)
	designation	VARCHAR(100)
FK	department_id	INT
FK	location_id	INT
	contact_no	VARCHAR(20)
	join_date	DATE
	employment_status	VARCHAR(20)

locations

PK	location_id	INT
	location_name	VARCHAR(100)
	location_type	VARCHAR(30)
	address	VARCHAR(255)

departments

PK	department_id	INT
	department_name	VARCHAR(100) UQ
	description	VARCHAR(255)

B · IT SUPPORT / TICKETING

ticket_tasks

PK	task_id	INT
FK	ticket_id	INT
FK	assigned_to_user_id	INT
	task_type	VARCHAR(50)
	description	TEXT
	status	VARCHAR(20)
	created_at	DATETIME
	completed_at	DATETIME NULL

tickets

PK	ticket_id	INT
	ticket_number	VARCHAR(20) UQ
FK	requester_employee_id	INT
FK	request_type_id	INT
FK	assigned_to_user_id	INT NULL
FK	linked_asset_id	INT NULL
	category	VARCHAR(30)
	priority	VARCHAR(20)
	subject	VARCHAR(200)
	description	TEXT
	attachment_path	VARCHAR(255)
	status	VARCHAR(20)
	deadline	DATETIME
	resolution_notes	TEXT
	submitted_at	DATETIME
	resolved_at	DATETIME NULL
	closed_at	DATETIME NULL
	updated_at	DATETIME

request_types

PK	request_type_id	INT
	type_name	VARCHAR(50) UQ
	applies_to	VARCHAR(20)
	description	VARCHAR(255)

C · ASSET & INVENTORY MANAGEMENT

assets

PK	asset_id	INT
	asset_tag	VARCHAR(30) UQ
	serial_number	VARCHAR(60) UQ
	asset_name	VARCHAR(100)
FK	asset_type_id	INT
	manufacturer	VARCHAR(60)
	model_number	VARCHAR(60)
	owning_company	VARCHAR(10)
	purchase_date	DATE
FK	vendor_id	INT NULL
FK	receiving_id	INT NULL
FK	current_location_id	INT NULL
	status	VARCHAR(20)
	condition	VARCHAR(20)
	warranty_expiry	DATE NULL
	notes	TEXT
	created_at / updated_at	DATETIME

asset_assignments

PK	assignment_id	INT
FK	asset_id	INT
FK	employee_id	INT
FK	assigned_by_user_id	INT
FK	returned_to_user_id	INT NULL
	assigned_date	DATE
	return_date	DATE NULL
	assigned_condition	VARCHAR(20)
	returned_condition	VARCHAR(20) NULL
	status	VARCHAR(20)
	remarks	VARCHAR(255)

asset_values

PK	asset_value_id	INT
FK	asset_id	INT
FK	value_type_id	INT
	purchase_value	DECIMAL(12,2)
	current_value	DECIMAL(12,2)
	accumulated_depreciation	DECIMAL(12,2)
	salvage_value	DECIMAL(12,2)
	valuation_date	DATE
	remarks	VARCHAR(255)

asset_value_types

PK	value_type_id	INT
	type_name	New / Old
	depreciation_percentage	DECIMAL(5,2)
	depreciation_method	VARCHAR(30)
	useful_life_years	INT
	description	VARCHAR(255)

asset_types

PK	asset_type_id	INT
	type_name	VARCHAR(50) UQ
	classification	Fixed / Consumable
	description	VARCHAR(255)

stock_movements

PK	movement_id	INT
FK	asset_id	INT NULL
FK	product_id	INT NULL
FK	performed_by_user_id	INT
	movement_type	VARCHAR(20)
	quantity	INT
	movement_date	DATETIME
	reference_number	VARCHAR(40)
	notes	VARCHAR(255)

asset_maintenance

PK	maintenance_id	INT
FK	asset_id	INT
FK	vendor_id	INT NULL
	issue_type	VARCHAR(50)
	description	TEXT
	service_date	DATE
	cost	DECIMAL(12,2)
	status	VARCHAR(20)
	resolution	TEXT
	returned_date	DATE NULL
	remarks	VARCHAR(255)

D · PROCUREMENT

demands

PK	demand_id	INT
	demand_number	VARCHAR(20) UQ
FK	product_id	INT
FK	requested_by_employee_id	INT
FK	department_id	INT
FK	location_id	INT
	quantity	INT
	justification	TEXT
	demand_category	VARCHAR(30)
	status	VARCHAR(20)
FK	created_by_user_id	INT
	created_at	DATETIME
	updated_at	DATETIME

products

PK	product_id	INT
	product_name	VARCHAR(100)
	category	VARCHAR(50)
	unit	VARCHAR(20)
	purchase_price	DECIMAL(12,2)
	current_price	DECIMAL(12,2)
	description	VARCHAR(255)
	is_active	BOOLEAN
FK	created_by_user_id	INT
	created_at	DATETIME

demand_approvals

PK	approval_id	INT
FK	demand_id	INT
FK	approver_user_id	INT
	decision	VARCHAR(20)
	approved_quantity	INT NULL
	comments	VARCHAR(255)
	approved_at	DATETIME
FK	created_by_user_id	INT

receiving_records

PK	receiving_id	INT
FK	demand_id	INT
FK	vendor_id	INT
FK	received_by_user_id	INT
	delivery_challan_no	VARCHAR(40) NULL
	gate_pass_no	VARCHAR(40)
	received_date	DATE
	received_quantity	INT
	condition	VARCHAR(20)
	remarks	VARCHAR(255)
FK	created_by_user_id	INT

vendors

PK	vendor_id	INT
	vendor_name	VARCHAR(100)
	contact_person	VARCHAR(100)
	contact_no	VARCHAR(20)
	email	VARCHAR(100)
	address	VARCHAR(255)
	tax_number	VARCHAR(30) NULL
	is_active	BOOLEAN
FK	created_by_user_id	INT
	created_at	DATETIME

invoices

PK	invoice_id	INT
FK	receiving_id	INT
	invoice_number	VARCHAR(40)
	invoice_date	DATE
	invoice_amount	DECIMAL(12,2)
	document_path	VARCHAR(255)
	status	VARCHAR(20)
	submitted_at	DATETIME
FK	created_by_user_id	INT
	created_at	DATETIME

payments

PK	payment_id	INT
FK	invoice_id	INT
	payment_reference	VARCHAR(40)
	payment_date	DATE
	paid_amount	DECIMAL(12,2)
	payment_method	VARCHAR(30)
	status	VARCHAR(20)
	remarks	VARCHAR(255)
FK	created_by_user_id	INT
	created_at	DATETIME

E · ONBOARDING & OFFBOARDING

onboarding_offboarding

PK	request_id	INT
FK	employee_id	INT
FK	request_type_id	INT
FK	initiated_by_user_id	INT
	request_date	DATE
	due_date	DATE
	status	VARCHAR(20)
	completed_at	DATETIME NULL
	notes	TEXT

checklist_items

PK	checklist_id	INT
FK	request_id	INT
FK	assigned_to_user_id	INT
FK	assignment_id	INT NULL
	task_name	VARCHAR(100)
	status	VARCHAR(20)
	due_date	DATE
	completed_at	DATETIME NULL
	notes	TEXT

F · AUDIT & SYSTEM CONTROL

audit_log

PK	log_id	INT
FK	user_id	INT
	action	VARCHAR(30)
	table_name	VARCHAR(60)
	record_id	INT
	description	TEXT
	created_at	DATETIME

LEGEND

PK Primary Key

FK Foreign Key

UQ = unique · NULL = optional value

1 — exactly one (parent)

0..1 — zero or one (optional)

0..N — zero or many (child)

1..N — one or many (child)

0..1 — zero or one (child)

Label at each line end gives the cardinality of that end of the relationship (parent → child).

DESIGN NOTES

1. Current holder of an asset = the asset_assignments row with no return_date; assignment history is never overwritten.
2. asset_types.classification = Fixed or Consumable (replaces the free-text asset type / category columns on assets).
3. asset_value_types (New / Old) hold depreciation %, method and useful life; asset_values stores purchase and current value.
4. products carry both purchase_price and current_price.
5. Every procurement table has created_by_user_id (who entered the record); approver / receiver columns record the business actor.
6. delivery_challan_no is optional (NULL allowed).
7. stock_movements: exactly one of asset_id / product_id is filled (CHECK constraint).
8. request_types serves ticket types and onboarding / offboarding / replacement types (see applies_to).
9. assets.receiving_id traces a fixed asset to its goods receipt; checklist_items.assignment_id links equipment items to assignments.

25 tables · 50 relationships · approx. 3NF